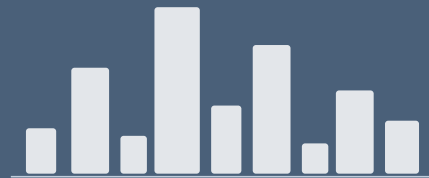


Interaction of a cumulus cloud ensemble with the large-scale environment, Part I

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How do a spectrum of cumulus clouds and the large-scale flow control each other, and how can the clouds' collective effect be predicted?



Matteo Di Sante

Paper review

Paper review: Arakawa & Schubert (1974) | Slide 1

Arakawa and Schubert build a closed theory of the mutual interaction between an ensemble of cumulus clouds and the slowly varying large-scale environment, including the subcloud mixed layer. This is the founding paper of the Arakawa–Schubert convection scheme, used in weather and climate models. Part I is theoretical: it derives the equations, introduces the cloud work function and the quasi-equilibrium closure, and checks that closure against tropical observations. Part II, announced in the paper as Schubert and Arakawa, gives the numerical solution method and applications. The towers on the cover are decorative: a spectrum of clouds with different tops.

Cumulus parameterization

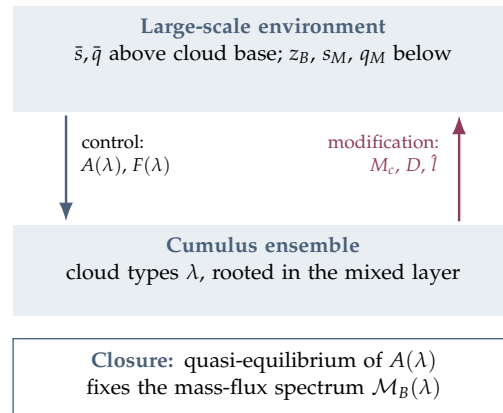
Many cumulus clouds live inside one large-scale disturbance, on much smaller scales. Predict only their collective effect, from large-scale variables.

Goal

A closed theory of the two-way interaction, with the subcloud mixed layer included.

Before this paper

CISK (Charney & Eliassen 1964) was too crude in general; Ooyama (1971) left the cloud spectrum open.



Closure: the clouds keep $A(\lambda)$ in balance with the large-scale forcing

The problem, and the road map

Cumulus clouds are much smaller and shorter-lived than the tropical disturbances that contain them. Models of the large scale cannot resolve them, so they need their collective effect as a function of resolved variables: this is cumulus parameterization.

Earlier theories. In CISK (conditional instability of the second kind; Charney and Eliassen 1964, Ooyama 1964) cumulus heating and the cyclone cooperate; the simple parameterizations gave insight and some success with tropical cyclones, but were too crude in general. Later schemes were empirical. Ooyama (1971) allowed clouds of different sizes, as non-interacting bubbles released from below, and showed that the problem reduces to a “dispatcher function” for the generation of bubbles; he left that function undetermined, so the scheme was not closed.

The diagram is the road map of the paper. Clouds modify the environment through their mass flux M_c , their detrainment D and the detrained liquid water $\hat{l}$ (Sections 2–3). The environment controls the clouds through the cloud work function A and the large-scale forcing F (Section 6). The cloud spectrum is built from entraining plumes labelled by λ (Section 4) that start from the subcloud mixed layer (Section 5). The closure (Section 7) is quasi-equilibrium of the cloud work function, originally proposed by Arakawa (1969), and it determines the mass-flux spectrum $\mathcal{M}_B(\lambda)$.

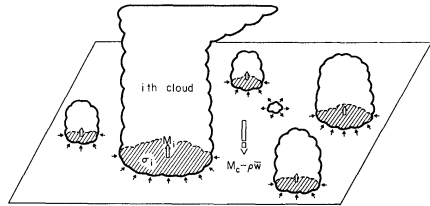


Fig. 1: a unit area between cloud base and the highest top. Tall clouds entrain; a cloud that lost buoyancy detrains.

$$M_i = \rho \sigma_i w_i, \quad M_c = \sum_i M_i, \quad \sigma_c = \sum_i \sigma_i \ll 1$$

$$s = c_p T + gz, \quad h = s + Lq$$

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L\hat{l}) + M_c \partial_z \bar{s} + \text{large scale}$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} + \text{large scale}$$

Detrainment: saturated cloud air with liquid water spreads out; its evaporation cools and moistens.

Cumulus-induced subsidence: warms and dries the environment between the clouds.

Condensation heating stays in the clouds. It keeps them buoyant and sustains M_c , so the environment warms through the subsidence, not directly.

The ensemble acts on the environment only through $M_c(z)$, $D(z)$ and $\hat{l}(z)$

How the clouds change their environment

Take a horizontal area large enough to contain many clouds but small compared with the large-scale disturbance. Cloud i covers a fraction σ_i and carries the mass flux $M_i = \rho \sigma_i w_i$. The large-scale mean vertical mass flux is $\rho \bar{w}$; the difference $M_c - \rho \bar{w}$ must go down between the clouds, so with active convection the environment subsides even where the large scale rises. Entrainment and detrainment follow from mass continuity for each cloud. A key simplification, supported by observations and theory (Bjerknes 1938; Asai and Kasahara 1967), is that the active clouds cover a small fraction, $\sigma_c \ll 1$. Then the environment's temperature and humidity practically equal the area averages, so predicting the environment is predicting the large scale. The variables are the dry static energy $s = c_p T + gz$, conserved in dry adiabatic motion, and the moist static energy $h = s + Lq$, conserved in moist adiabatic motion.

The two budget equations, (74) and (75), are the core of Sections 2–3. In teal, detrainment: at the level where a cloud loses buoyancy, saturated air with liquid water spreads into the environment, and evaporation of that water cools and moistens. In red, cumulus-induced subsidence, which warms and dries a stably stratified environment, since $M_c \partial_z \bar{s} > 0$ and $M_c \partial_z \bar{q} < 0$. In grey, the large-scale terms: advection, mean vertical motion and radiation.

The important physical point, used first by Arakawa (1969), is that latent heat released inside the clouds does not warm the environment directly: it maintains the buoyancy against entrainment and adiabatic cooling, hence the cloud mass flux, and the environment warms through the compensating subsidence. The vertical distributions of heating and condensation can therefore be very different. So the problem reduces to $M_c(z)$, $D(z)$ and the detrained liquid water $\hat{l}(z)$.

A spectrum of entraining plumes

Cloud type λ : fractional entrainment rate, constant with height. Since $\mu = 2\alpha/R$, large λ means small clouds.

$$\partial_z \eta = \lambda \eta \Rightarrow \eta(z, \lambda) = e^{\lambda(z-z_B)}$$

$$h_c(z, \lambda) = \frac{1}{\eta} \left[h_M + \lambda \int_{z_B}^z \eta \bar{h} dz' \right]$$

Top $z_D(\lambda)$: buoyancy vanishes, $h_c = \hat{h}^* \approx \bar{h}^*$.

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

Given the environment, each cloud type is fixed; only $\mathcal{M}_B(\lambda)$ is unknown

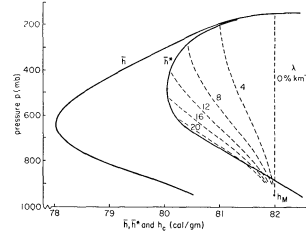


Fig. 6: $\bar{h}$, $\bar{h}^*$ and h_c for $\lambda = 0-20\% \text{ km}^{-1}$; tropical sounding.

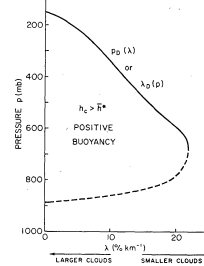


Fig. 8: top $p_D(\lambda)$.

In the paper

The clouds start from the subcloud mixed layer, $h_c(z_B, \lambda) = h_M$, whose depth z_B is predicted and is itself part of the control on convection.

The total detrainment at a level is the mass flux of the clouds that lose buoyancy there. This suggests dividing the ensemble into sub-ensembles by cloud type. One parameter is assumed to be enough. The authors choose the fractional entrainment rate λ of the time-averaged mass flux, assumed constant with height. In Simpson's one-dimensional cumulus-tower model the entrainment rate is $2\alpha/R$, with R the tower radius, so a large λ means a small, strongly diluted cloud and a small λ a large one. A cloud need not be a steady jet: it can be a sequence of active elements with little horizontal expansion, and a constant radius in the Lagrangian sense fits observations better than expanding thermals or starting plumes (Simpson et al. 1965).

With constant λ , the normalized mass flux η grows exponentially from the base z_B up to the detrainment level $z_D(\lambda)$, where it drops to zero. The moist static energy in the cloud is the mixed-layer value diluted by entrained environmental air. The cloud top is where the cloud's buoyancy vanishes, approximately where h_c meets the saturation moist static energy $\bar{h}^*$ of the environment. Figure 6 shows this for a mean tropical sounding: the dashed h_c curves meet $\bar{h}^*$ lower for larger λ . Figure 8 is the resulting top pressure: smaller clouds detrain lower. The total mass flux is an integral of $\mathcal{M}_B \eta$ over the types still rising at z , and the detrainment at z comes from the type whose top is at z .

Section 5 gives the mixed layer. Observations over the Caribbean show a well-mixed layer between the sea surface and a level about 200 m below cloud base, in which s , q and h are nearly constant, with jumps at its top z_B . The clouds are assumed to start with mixed-layer properties. The depth equation (126) balances the subsidence between the clouds, which pushes z_B down, against turbulent entrainment driven by the surface buoyancy flux, with a coefficient k of about 0.10 to 0.25. A shallow mixed layer feeds a smaller share of mixed-layer air to the clouds, so its depth is one of the controls on the intensity of convection.

So, once the environment is known, every thermodynamic property of every sub-ensemble is known, and the only missing quantity is the mass-flux distribution function $\mathcal{M}_B(\lambda)$ at cloud base.

The cloud work function and the closure

$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \frac{g}{c_p \bar{T}} \eta(z, \lambda) [s_{vc} - \bar{s}_v] dz$$

Kinetic energy generated per unit cloud-base mass flux, fixed by the environment alone. $\lambda = 0$: undiluted ascent, insensitive to the humidity above cloud base; larger λ also needs a moist environment.

$$\frac{dA}{dt} = \underbrace{\int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'}_{\text{clouds, } K < 0} + \underbrace{F(\lambda)}_{\text{large scale}}$$

Clouds consume their own instability; the large scale rebuilds it, mainly by cooling through ascent.

Quasi-equilibrium

$\tau_{ADJ} \sim 10^3\text{--}10^4$ s, the time the clouds need to remove A , against $\tau_{LS} \gtrsim 10^5$ s for the forcing. Then dA/dt is negligible:

$$\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) = 0$$

A Fredholm equation of the first kind for $\mathcal{M}_B \geq 0$: the spectrum of clouds is the one that balances the forcing.

In the paper

A sequence of quasi-neutral states, not a steady state: A stays small, not constant.

The clouds adjust fast enough that the large scale fixes their spectrum

Section 6 asks how the large scale controls the clouds. The kinetic energy of type λ changes by generation minus dissipation, and generation is proportional to the cloud-base mass flux, with coefficient $A(\lambda)$: the cloud work function, the kinetic energy generated per unit mass flux. It is the buoyancy of the cloud air, weighted by the normalized mass flux η and integrated from the mixed-layer top to the cloud top.

Because h_c and the cloud top are fixed by the environment, $A(\lambda)$ is a property of the environment alone, mixed layer included, and the only one that can influence kinetic-energy generation by type- λ clouds. Rewriting it shows buoyancy at cloud base plus conditional instability, reduced by a term proportional to λ and to $\bar{h}^* - \bar{h}$: non-entraining clouds do not feel the humidity above cloud base, strongly entraining ones need a moist environment. So $A(\lambda) > 0$ is a generalized, cloud-type-dependent criterion for moist convective instability. It is necessary, not sufficient: generation also needs $\mathcal{M}_B > 0$ and a trigger. For $\lambda = 0$ the integral is close to what is now called CAPE.

The tendency of A splits into cloud terms, linear in the mass-flux distribution with kernel $K(\lambda, \lambda')$, and the large-scale forcing $F(\lambda)$. The kernel is typically negative: clouds use up their own instability, mostly through the subsidence warming induced by the other types. Its detrainment part is positive for $\lambda' > \lambda$, since shallower clouds moisten and cool the environment of deeper ones. The forcing is dominated by cooling of the cloud layer by large-scale ascent, plus the deepening of the mixed layer.

Order of magnitude: the cloud term is A/τ_{ADJ} , the actual tendency at most A/τ_{LS} . With a cloud buoyancy of a few degrees, they estimate $\tau_{ADJ} \sim 10^3\text{--}10^4$ s against $\tau_{LS} \gtrsim 10^5$ s, so the tendency can be dropped and the cloud term balances the forcing. That is Eq. (150), a Fredholm integral equation of the first kind for $\mathcal{M}_B$. Since $\mathcal{M}_B$ cannot be negative, where the formal solution would be negative the type is absent and the balance becomes an inequality. Quasi-equilibrium does not mean steady temperature and humidity: it is a sequence of quasi-neutral states in which A is small, of order τ_{ADJ} times the forcing. Only the combination $\mathcal{M}_B = \mathcal{N} m_B / \tau$ is needed, not the population, the lifetime and the single-cloud mass flux separately.

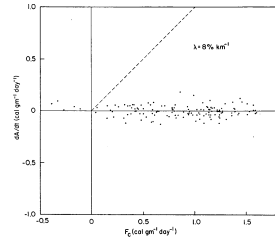


Fig. 13: observed dA/dt against the forcing F_C , Marshall Islands. Dashed: the change without clouds.

Observational check

$dA/dt \approx 0$ while F_C reaches $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$: the clouds consume the instability as fast as it is made.

A closed statistical theory of convection, tested on one tropical data set

Conclusions

The ensemble acts on the environment only through M_C , D and $\hat{I}$.

Split into entraining plumes, each cloud type is fixed by the environment; only $M_B(\lambda)$ is left.

$A(\lambda)$ measures moist convective instability for each cloud type.

Quasi-equilibrium closes the theory: $M_B(\lambda)$ solves a Fredholm equation of the first kind.

Limits

One parameter λ per cloud type; entraining updrafts only, no downdrafts.

Detrainment only at cloud top; crude rainfall; $\sigma_c \ll 1$; horizontally uniform environment; unsaturated mixed layer.

Only $M_B(\lambda)$: no population, lifetime or cloud size. Triggering assumed, not modelled; shear neglected.

One tropical oceanic data set, without F_M ; the solution method is left to Part II.

Conclusions and limits

Since $A(\lambda)$ depends only on the thermodynamic profiles, it can be computed from radiosondes and its tendency from time series; the forcing needs wind observations over a network, here the Marshall Islands data of Yanai et al. (1973). If the clouds did nothing, the points of Figure 13 would lie on the dashed line $dA/dt = F_C$. Instead the observed tendency scatters around zero while the forcing ranges up to about $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$, for both cloud types shown. Figure 14 of the paper adds that the two parts of the work function swing by about 20 cal g^{-1} over ten days at Eniwetok, while their difference, A , stays small and positive.

Conclusions. The cumulus ensemble modifies the environment only through the total cloud mass flux, which drives the compensating subsidence, the total detrainment, and the liquid water detrained at the level of vanishing buoyancy. Represented spectrally, by the fractional entrainment rate, each sub-ensemble is completely determined by the environment once the conditions at the top of the mixed layer are known, so the parameterization reduces to the mass-flux distribution function. The cloud work function is a property of the environment and a generalized, cloud-type-dependent measure of moist convective instability. Under quasi-equilibrium, the mass-flux distribution must satisfy a Fredholm integral equation of the first kind: this is the principal conclusion of Part I.

Limits. A single parameter is assumed to characterize a cloud type, and the clouds are entraining updrafts, with no convective-scale downdrafts. Detrainment happens only in a thin layer near the top, and rainfall is parameterized very crudely. The active clouds must cover a small fraction of the area, the environment is horizontally uniform, and only an unsaturated mixed layer is treated. The theory predicts only the mass-flux distribution, not the number of clouds, their size or their lifetime, and these are not equivalent. A trigger is needed to start convection, but it is not modelled, and mechanical interactions with the vertical shear are neglected. On the evidence side, the quasi-equilibrium check uses one tropical oceanic data set and omits the mixed-layer forcing, and the solution of the integral equation is deferred to Part II.